

Text and Dynamic Network Analysis for Measuring Technological Convergence: A Case Study on Defense Patent Data

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Abstract—Identifying technology convergence is getting harder, especially in fast-evolving and cross-technological fields. On the other side, natural language processing and network analysis researchers is to provide a novel method for mapping technologies and their relations over time, in order to identify dynamic patterns of convergence and to test it on the C4ISTAR field, a defense-related cross-technological field. The methodology automatically extracts technologies from a corpus of scientific papers, policy documents, websites, and company reports using Named Entity Recognition approaches. These technologies are then retrieved from the second corpus of more than 300 thousand patents related to the C4ISTAR domain, measuring their cooccurrences over time. Finally, using the time-varying network analysis, we were able to identify and measure the pattern of technological convergence.

Index Terms—Defense sector, natural language processing (NLP), patents analysis, technological convergence, text mining.

I. INTRODUCTION

TECHNOLOGICAL convergence describes the trend toward a merge of distinct technologies into a unified whole [1]. Studying this phenomenon is particularly important today, considering that it is already taking place in a number of industrial fields [2], [3], opening up new business and growth

opportunities. Despite its relevance, academic literature providing insights into convergence is rather scarce [4] and not able to take advantage of the opportunities that recent trends in artificial intelligence are offering to this field of study on the analysis of big unstructured datasets [5].

In this article, we examine technological convergence by developing a novel methodology that combines text mining and dynamic network models. The methodology is applied to more than 300 000 patents and exploits Named Entity Extraction techniques to identify technologies.

The focus on patents as semantic data sources is driven by three arguments: 1) it is widely accepted that 70%–90% of the (technical) information in patents is not published elsewhere [6]; 2) patents are technologically focused documents, thus they are more likely to contain technologies than other documents such as scientific papers [7]; 3) patent documents are published with a large quantity of metadata. The usage of metadata allowed us to create technological networks and trace their evolution over time.

The method is applied on a real-world case study in the context of C4ISTAR, an acronym used in Defense for *Command, Control, Communication, Computer, Information/Intelligence, Surveillance, Target Acquisition, and Reconnaissance*. It is a broad concept that states the functions of military missions and for the systems that implement actions and coordination [8]. We test our method on a sector that has been impacted by two important convergence effects. First, digitalization with the growing interplay between conventional and digital technologies. Second, an increasing effort to use civilian technologies and to pursue dual-use to cut development costs [9].

To sum up, we contribute to the state of the art in three ways. First, we develop a novel methodology for text-driven profiling of a technological field that combines heterogeneous textual sources. Second, we combine text mining with dynamic networks. Third, the method is tested in the C4ISTAR domain, a large, complex, and ever-evolving defense field.

The rest of this article is organized as follows. Section II describes the C4ISTAR field and Section III reviews the theoretical work on document retrieval, technology extraction, and technological convergence. In Section IV, we describe the method adopted for technology retrieval, the construction of patent corpus, and network analysis. Section V presents the results. Finally, Section VI concludes this article.

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II. C4ISTAR FIELD

C4ISTAR is the outcome of the progressive integration of information technologies in the field of military operations. Given the variety of operational conditions and the turbulent dynamics of both defense and digitalization, the C4ISTAR field has become, in a few decades, a vast and complex technological field, mainly for two reasons.

First, C4ISTAR has fuzzy boundaries. It comes from the concept of Command and Control (C2), a set of organizational and technical attributes that employ human, physical, and information resources to accomplish missions [8]. In the C2 literature, it is difficult to identify a common definition of C4ISTAR. There is indeed a large set of definitions each of which is focused on a particular perspective [10]. A useful starting point to understand in detail the concept is to examine each subfield of the C4ISTAR acronym.

- 1) *C2* includes all functions necessary for planning operation.
- 2) *Communication and Computer* provide secure communication means.
- 3) *Intelligence and Information* are referred to the knowledge and data necessary to accomplish missions.
- 4) *Surveillance and Reconnaissance* are obtained by systematic observation and detection methods.
- 5) *Target Acquisition* is related to the detection, identification, and location of a target.

However, the term C2 has been coupled with a plethora of subfields over time. Evidence is provided in Appendix A, where we list the acronym variants, reported in the Wikipedia page *Command and Control*.¹

Second, it is a defense-related domain. Defense is the set of systems and methods used to protect the integrity of state territory, the security of its population, and the maintenance of civil institutions. In every nation, there is a Defense Industrial Base (DIB), a group of firms that provides defense-related equipment to the Defense Ministry. The Government supports its DIB and it is also its main consumer. As a result, the Defense Industry is shaped by national policy and by government financial decisions [11].

Identifying technological convergence is a well-known issue in the defense domain. Russel *et al.* [12] analyze the Internet of Battlefield things and discuss the implications of their technological convergence on military decision making, while Trimmer [13] studies the convergence phenomena in Radar and Electronic Warfare.

III. THEORETICAL BACKGROUND

Before addressing the issue of technological convergence, it is necessary to apply proper technological delineation techniques and to control for the quality of information retrieval (IR) in order to avoid a garbage-in-garbage-out effect. Given that C4ISTAR is particularly complex, the present section analyses current approaches and open challenges for patent retrieval, technology extraction, and finally, convergence analysis in order to have the proper theoretical (and practical) basis for our method.

A. Patent Retrieval

The common strategies to retrieve patents from a database are based on keywords [14] patent classification systems (e.g., International Patent Classification, also known as IPC²) [15] or their combination [16]. Both methods have advantages and drawbacks, as explained by Bruns *et al.* [16]. For this article, we cannot exploit classification systems due to the fact that these are not designed to enclose domains that combine different technological subfields, as in the case of the C4ISTAR [17].

A query of keywords is defined as a structured sequence of words, connected by logical units (e.g., OR, AND, NOT) to be searched in a database. In general, this initial set of keywords are provided by a panel of experts [18] in the analyzed domain. However, the expert-driven approach suffers from several limitations [19]: they are time consuming, subject to significant time constraints (since experts are famous to be rarely available) [20], lack systematic rules [21], and are bias prone [22]. These limitations have become more evident over the last two decades after several efforts of many authors to reliably delineate the field of Nanotechnology. Most studies indeed provided largely different field delineations of the emerging Nanotechnology domain [15].

The phase of document collection is crucial in order to avoid the garbage-in-garbage-out effect. For this reason, we described our approach of patent retrieval in particular details in Section IV-A.

B. Technology Extraction

Once a set of documents has been collected, there exists the need to extract technological information in order to create a network and trace convergence. This may take place using the citation approach, lexical approach, or mixes of the two.

Since the ground-breaking works of Garfield on citation indexing [23], citation networks have been widely used [24]. Over the last decade, there was also advancement in hybrid approaches that combine citation-based and term-based techniques [25]. In turn, lexical approaches construct networks using textual data [26]. The most used approaches are term-based probabilistic methods [15].

In this article, we focus on another technique for technology mapping, NER, an Information Extraction technique for recognizing unstructured text information units (e.g., foods, person names, companies, geographical entities) [27].

The NER methods used in the literature are three. First, *gazette-based NER* (also known as *terminological-based*). It aims to map mentions of entities within texts to terminological resources (e.g., Wikipedia) [27]. It is a fast and accurate approach to NER, that uses a list (or gazette) consisting of known instances [28]. It is defined as semisupervised since Gazette creation requires manual effort and domain-specific expertise. The list is then automatically expanded searching for semantically similar entries (using word representations such as BERT) and automatically searched in the documents. The second approach

¹[Online]. Available: https://en.wikipedia.org/wiki/Command_and_control, Accessed: Mar. 4, 2021.

²[Online]. Available: <https://www.wipo.int/classifications/ipc/en/>, Accessed: Mar. 4, 2021.

is *rule-based NER*. It uses lexicons, regular expressions, and lexical information to express knowledge-based systems able to extract a certain type of entity [29]. These are unsupervised top-down approaches, also referred to as Expert Systems. The third approach is, *corpus-based NER*: uses manually tagged text corpora (well known in the literature as the training set, or annotated documents) to train machine learning algorithms [30]; these techniques are at the state of the art but need a huge manual effort.

The main existing metrics to measure the performance of IR [31] and NER systems are Precision, Recall, and F-measure [32]. Tsai *et al.* [32] define Precision as the number of named entities (NEs) that a system correctly detected in textual data divided by the total number of NEs identified by the system. Recall is the number of NEs a system correctly detected divided by the total number of NEs contained in the input text. F₁-Score combines these two metrics into their harmonic mean.

In general, state of the art is achieving good results for the recognition of NEs. Recently, Lee *et al.* [33] reached state of the art in recognizing NEs in the context of biomedical entities, obtaining an f1 score of 89.61 for diseases, 94.26 for drugs, and 78.58 for Gene and Proteins names. Similarly, a level of f1 of 75.37 has been reached in the extraction of events and location entities [34]. Both techniques used contextual word embedding (i.e., BERT) on top of deep neural networks as described in the seminal paper by Devlin *et al.* [35].

To the best of our knowledge, the literature on NER systems for extracting technologies is poor. The only similar work we were able to retrieve is a preprint by Hossari *et al.* [36] that focuses on the extraction of Artificial Intelligence technologies and gives qualitative criteria for evaluating the results. Other works extract only emerging technologies and not established ones, exploiting IPC classes [37] or using full text but with greedy approaches, extracting generic terms and not only technologies [38].

This gap in the literature is due to the fact that while a training set exists for entities such as chemical molecules³ (on which is possible to train a corpus-based classifier), this is not true for technologies. This is the reason why in this article we cannot use state-of-the-art systems (i.e., BERT) but we rely on unsupervised methods, generally reaching lower performances in terms of precision. Anyway, even if this limits the efficiency of the extraction, it is important to notice that the precision does not impact the quality of the output (effectiveness of the whole process) since, in this article, we manually review the list of technologies, in order to filter nontechnologies entities and to offer a baseline in terms of precision for further works.

For what concerns recall, the lack of a training set force to evaluate it considering the grain of the extracted entities. Linguistically, we can refer to technologies using abstract wording (e.g., *device* or *system*) or specific wording (e.g., *waterproof smartwatch* or *deep recurrent neural networks*). In deciding the correct level of grain for measuring convergence, it is important to avoid both the collection of too generic and too specific technologies.

For what concerns the risk of too generic technologies, we avoid it by using text-based extraction. From our review of the literature, related works stop the analysis at a coarser grain, such as in the case of IPC classes [39]. In general, considering too generic technologies arises the risk that no dynamics would be identified by the network analysis. With respect to a too fine grain, we have to avoid collecting subsystems instead of technologies. To clarify, let us consider the case of *satellite images*, a typical technology of C4ISTAR. Considering a higher level of grain would mean to collect more technologies (have higher recall and collect subtechnologies of *satellite images*) but also to switch the focus from C4ISTAR to the field of satellite images, going out of scope for the analysis.

For this reason, the list of technologies and the rule-based extraction avoid the risk of falling into these problems, making it possible to use text analysis for measuring technological convergence.

C. Technological Convergence

The first use of the term “convergence” is attributed to Rosenberg [40], who used it to describe processes from unrelated industries to solve their technical problems. Subsequently, convergence has been referred to as the blurring of boundaries between industries through the convergence of elements such as value propositions, technologies, and markets [41], as the innovations that emerge at the intersections of clearly defined industry boundaries, producing a much broader impact [42] and as the combination of existing technologies that also aims to create innovations not previously in existence [43].

Previous research on technology convergence in different fields demonstrated that the phenomenon arises from the transfer of scientific achievements to industry [44]. For example, science convergence, mainly related to interdisciplinary and transdisciplinary research, leads to the emergence of new technologies as well as new markets [45].

Consequently, when the push from technological convergence is associated with the pull from market convergence, we may witness the emergence of new industries or large segments of industries [46].

When a convergence dynamic takes momentum, companies are challenged in their strategies. From an external strategic-based perspective, convergence increases the strategic uncertainty because it is not clear who might take the leadership of the process and establish a competitive advantage. From an internal resource-based perspective, the main uncertainty refers to the ability of incumbent companies to identify the relevant external technologies to manage them and to master the related skills. Here, text mining has recently been demonstrated to be a suitable method to face this growing complexity [47].

The complexity of convergence dynamics has been recently increased by Covid-19. The first approach to detect weak signals of the convergence that occurred during the pandemic has been addressed by Melluso *et al.* [48]. The method has a rationale that is similar to the one adopted in this article (a combination of text mining and network analysis but consider a different input (medium blogging platform instead of patents), for a different purpose (detect emerging technologies instead of convergence)

³[Online]. Available: <https://www.ebi.ac.uk/chebi/>, Accessed: Mar. 4, 2021.

TABLE I
STATE OF THE ART OF PREVIOUS APPROACHES TO THE DETECTION OF
TECHNOLOGICAL CONVERGENCE USING PATENT DATA

Data	Technique	References
Citations IPC-classes	Network Analysis	[49], [50], [51]
	Cluster Analysis	[53]
	Topic Modeling	[54]
Text	Network Analysis	[58]
	Keywords Extraction	[59]
	Bi-gram extraction	[56]
	Sentiment Analysis	[57]
	Topic Modeling	[55]

using different methods (static network analysis instead of dynamic networks analysis).

Furthermore, it is interesting to notice that most methods developed for measuring technological convergence focused on patents. Table I summarizes the most important research contributions in technological convergence pattern identification that use patents. This table divides related works (column 3) based on the focus of the analysis in terms of input information (column 1) and the used technique (column 2).

First, studies that focused on citation analysis used network analysis techniques to identify technological convergence. Karvonen *et al.* [49] provided a tool for anticipating the early stages of convergence using citation networks analysis. Cho *et al.* [50] identified strategy transformation under technology convergence in the printed economic industry. Kim *et al.* [51] proposed a forecasting methodology for multitechnology convergence, based on a patent-citation analysis.

Second, studies that focused on IPC classes used different techniques. Lee *et al.* [52] proposed a new approach for clustering technologies based on their growth patterns analyzing patent data using a hidden Markov model. Gauch and Blind [53] used IPC classes to identify convergence trends using a revised version of a cross-impact assessment method. Song *et al.* [39] introduced an IPC-network classification method for the anticipation of converging technology areas based on the concept of knowledge flow. However, IPC classes were further analyzed by Lee *et al.* [54] with a text-based technique, called latent Dirichlet allocations, to detect emerging areas and technology convergence.

Third, we have research works that shifted the focus toward the analysis of the textual content of the patents. In particular, Venugopalan *et al.* [55] proposed topic models to classify patents based on their technological convergence stage. Eilers *et al.* [56] extracted bigrams from text to capture one-way and two-way technology convergence. Zhou *et al.* [57] used sentiment analysis to evaluate technological candidates for possible future recombination.

Finally, several authors focused on combining citations, IPC classes information, and textual content with the use of different techniques. For example, Sasaki and Sakata [58] proposed a mixed methodology to identify technological spin-offs combining patent citation analysis and IPC association analysis. Kim *et al.* [59] used the abstracts of the patents to extract keywords and performed a network analysis of the IPC classification to detect

TABLE II
SEARCH STRATEGY FOR RETRIEVING THE PATENT DOCUMENTS ON THE
C4ISTAR FIELD

Search Strategy	Terms in OR
C4ISTAR related terms	<i>c4istar</i> , <i>surveillance</i> , “ <i>target acquisition</i> ”, <i>reconnaissance</i> , <i>command&control</i> , <i>communication</i> , <i>intelligence</i> , <i>command</i> , <i>control</i>
Defense related terms (AND)	<i>military</i> , <i>defense</i> , <i>defence</i> , <i>weapon</i> , <i>battleship</i> , <i>battlefield</i> , <i>battlespace</i>
Black list (AND NOT)	<i>Neutrophil</i> , “ <i>lipid peroxidation</i> ”, <i>immunology</i> , <i>protein</i> , <i>upregulation</i> , “ <i>superoxide dismutase</i> ”, <i>catalase</i> , <i>phagocytosis</i> , <i>metabolism</i> , “ <i>oxidative stress</i> ”, <i>inflammation</i> , “ <i>immune response</i> ”, <i>pathogenesis</i> , <i>macrophages</i> , <i>infection</i> , <i>cytokines</i> , <i>glutathione</i> , <i>macrophage</i> , “ <i>nitric oxide</i> ”, <i>antioxidant</i> , <i>pathology</i> , <i>cytokine</i> , <i>pathophysiology</i> , <i>epidemic</i> , <i>microbiology</i> , <i>enzyme</i> , <i>disease</i> , <i>immunity</i> , <i>genetics</i>

the dynamic evolution of key technologies in printed electronics technology. Kim *et al.* [60] performed technology opportunity discovery by extracting representative keywords from the text and compare the results with the citation networks of the patents analyzed. To sum up, the current scientific literature lacks combined approaches that use network analysis directly on textual data, exploiting technology collection techniques as NER. The proposed method is novel in that sense since it combines textual and network analysis on patents to detect patterns of technological convergence. The novelty of the method and its characteristics are discussed in the next session.

IV. DATA AND METHODOLOGY

In this section, we show the methodological steps undertaken to identify and analyze the technological convergence (or divergence) of the C4ISTAR domain. The methodology is structured in three macrotasks: 1) constructing a query for retrieving a patent set on C4ISTAR domain; 2) extracting technologies from a collection of the C4ISTAR documents to have a list of technologies adopted in C4ISTAR; 3) modeling and analyzing the technological convergence using the time-varying network analysis. In the following sections, we describe in detail each step.

A. Patent Retrieval

The keywords search strategy starts with the selection of terms that allows retrieving all and only C4ISTAR related patents. In our case, we selected an initial sequence using the subfields included in the C4ISTAR acronym linked with a set of words referred to the defense sector by the AND logical operator. Table II provides the terms related to these two search strategies (see C4ISTAR and Defense-related terms' rows in Table II).

The query allows us to retrieve 347 589 patent documents on the Erre Quadro S.R.L. database.⁴ The database contains over

⁴[Online]. Available: <https://www.errequadrosrl.com/>, Accessed: Mar. 4, 2021.

90 million patents and includes high-quality bibliographical and legal status patent information from leading industrialized and developing countries. It indexes patents from different European and International databases of patents (e.g., EPO, WIPO).

Considering the ill-defined nature of C4ISTAR, in order to improve the quality of the output, we constructed a list of elements that should not be included in the Title, Abstract, or Claims section of the patents. The exclusion of terms in a query for improving the result of the retrieval process is well established in the literature. For example, Porter *et al.* [61] and Mogoutov and Kahane [62] use a list of words to remove nonrelevant keywords in their search on Nanotechnologies. We referred to a list of terms that must be excluded as a blacklist. The blacklist approach has been designed for the specific case of C4ISTAR and contains terms from the medical and from the gaming field. More information on this approach is in Appendix B.

The blacklist approach proposed in this article provides information about how we detected and evaluate the boundaries of C4ISTAR: identifying what is similar to the domain but not inside the domain made it possible to design a better query. This process was essential to map and measure the convergence phenomenon avoiding the garbage-in-garbage-out effect.

Consequently, the new query is defined by the C4ISTAR related terms, the defense-related terms, and the items in the blacklist, combined as given in Table II.

The query was performed in May 2020 and retrieved 306 213 patents; it means that 41 376 are removed from the 347 589 initial sets.

In order to evaluate the quality of the patent retrieving process, we measured the precision of our query. From our review of the literature, an established sample dimension for evaluating precision is 1% of the patent set [15]. For this reason, a set of 3063 patents has been randomly sampled from the patent set. Then, the authors revised the patents evaluating the title and the abstract of the patents. The assignment for the group was: "Identify whether an analyzed patent is into the scope of the C4ISTAR or not." As a result, 298 patents were labeled as false positive having a precision of 0.90.

As a robustness check, we calculated the confidence intervals of the precision based on the Wilson method [88] with a confidence coefficient of 95%. The method provided an upper bound of the precision of 0.9127% and a lower bound of 0.9017%. This result is in line with similar works [16].

B. Technologies Collection

In this section, we show how we collected C4ISTAR technologies. The technologies will feed a dictionary that will then be used to implement the gazetteer-based search. This approach, together with the rule-based approach, is how we implemented the technology NER system.

1) *Data Collection and Preprocessing*: The technology collection starts with the retrieval of heterogenous documents related to the C4ISTAR field. To reduce the bias effect of a single source and to increase the recall of the technological mapping, we use different types of sources for the document collection. In

fact, the official documents by national or international institutes may be influenced by political decisions; company documents are written based on the main competencies and products of the company; online news on C4ISTAR may offer only a single perspective. For these reasons, we used the following five sources. First, *National or international institutions report*: 12 documents by governments and public services with an institutional purpose [63]. The second source is *Research institutions documents*: 23 scientific publications related to the C4ISTAR area, collected from the Scopus database and from the C2 and Cyber Research Portal [64]. Third, *Corporate documents*: 72 among annual reports, press releases, technical fiche, or datasheet describing products related to C4ISTAR systems. The selection of companies related to the C4ISTAR is based on their appearance in SIPRI TOP 100.⁵ Fourth, *Thematic website news*: 58 Internet news about the C4ISTAR field, extracted from C4ISRNET.⁶ We retrieved a set of 166 documents published from 2000 to 2020 (166 documents) in the English Language.

The collected documents are then preprocessed. The first preprocessing step is tokenization, in order to subdivide the text into single units of meaning, also called token. A token could be a single word, a bigram (a pair of words), a trigram, or a chunked word. A chunked word is a multiword with its own meaning different from the meaning of words that composed the multiword. The second step is part-of-speech tagging (or PoS tagging) to assign to each token the part of speech. The PoS tagging is a Natural Language Processing (NLP) technique for assigning unambiguous grammatical categories to words in context [65].

The preprocessing steps were implemented using the R package "udpipe" [66] for performing the PoS tagging process that implements the CoNLL-U model.

After the preprocessing steps, the Named Entity Recognition technique allows us to automatically identify the C4ISTAR technologies. The NER methods used for this purpose are a gazetteer-based approach and a rule-based approach.

2) *Gazetteer-Based Method*: For the gazetteer-based approach, we automatically listed relevant entities that are collected in a gazette (a dictionary or a list of terms). In our case, the relevant entities are the technologies and the gazette is created starting from several predefined lists of technologies. The sources, freely available online, are the lexicon developed by Chiarello *et al.* [22] containing 1320 technologies, O*NET,⁷ a database developed for the U.S. Department of Labor made of 974 occupations and their corresponding skills, knowledge, abilities, and technologies (about 30 000), Wikipedia list of emerging Technologies⁸ and Techopedia,⁹ a database of 2000 technologies, widely considered by experts in the IT sector as a go-to tech source for technologies.

⁵[Online]. Available: <https://www.sipri.org/databases/armsindustry>, Accessed: Mar. 4, 2021.

⁶[Online]. Available: <https://www.c4isrnet.com/>, Accessed: Mar. 4, 2021.

⁷[Online]. Available: <https://www.onetonline.org/>, Accessed: Mar. 4, 2021.

⁸[Online]. Available: [List_of_emerging_technologies](#), Accessed: Mar. 4, 2021.

⁹[Online]. Available: <https://www.techopedia.com/dictionary>, Accessed: Mar. 4, 2021.

These lists of technologies are merged together in a unique gazette and the occurrences of all these technologies are identified in the C4ISTAR documents.

3) *Rule-Based Method*: The rule-based approach aims to define a list of semantic rules for retrieving the technologies from the C4ISTAR documents. The rules constructed are two. First, technology is a noun chunk. A noun chunk is composed of a sequence of (*NOUNs or ADJECTIVEs*) + *NOUN*. For example, *machine learning* is a noun chunk composed of *machine* and *learning*. Second, the last noun of the technology noun chunk is one of the following terms: *machine*, *technology*, *device*, *apparatus*, *mechanism*, *sensor*, *network*, and *system*. This list of the terms depends on the domain that we are analyzing (it can be adapted for extracting technologies from documents of different domains). Our list is constructed considering the work of Jang *et al.* [67]. They defined the words *system*, *unit*, *device*, *apparatus*, and so on as the high-level of generic concepts in terms of automotive technology. We expand this set of words for applying their findings in our domain, as described above.

4) *Similarity Metrics*: The technologies are identified, and similar entities are clustered automatically using similarity metrics, i.e., Levenshtein and Jaro–Winkler [68]. After a series of empirical tests, we chose the threshold value of 2 for Levenshtein distance and 0.15 for the Jaro–Winkler metric. Therefore, if this distance is below the thresholds, we assume a match. Other well-known algorithms for detecting similarity are using contextual word embedding, i.e., the BERT model [35], which however are hard to adapt to our domain since as discussed in Section III, there is no training set for technologies.

Finally, the dictionary is manually cleaned by two Master’s students in engineering management and a Ph.D. student in the smart industry at the University of Pisa, in a double-blind mode. We provided both students with a table containing the list of extracted entities (the same for both students). The assignment was “Read each extracted entity and decide whether the entity is a technology or not.” Students were allowed to use external sources for achieving their goals. The tables of the results are then compared and if there was disagreement, the students start a debate and converge to a common solution. The manually cleaned statement aims at calculating the precision of our NER system in the identification of technologies task. The result of this step is shown in Section V-A (see Table III).

For reproducibility, the final list of technologies is available as open data [69].

C. Technological Convergence Analysis

This section discloses the steps for detecting technological convergence. Each section describes a specific phase. The overall analysis described in the following sections was conducted using Spacy for NLP.¹⁰

1) *Technologies Extraction*: The list of technologies collected according to the method described in Section IV-B is retrieved from the patent set gathered using the query strategy defined in Section IV-A.

TABLE III
RESULTS OF THE NER PROCESS FOR EXTRACTING TECHNOLOGIES

Approach		N° of technologies	Precision	Entities Extracted (Random sample of 5 technologies)
Gazetteer-based	O*NET	162	74.65 %	acoustic sensors; artificial intelligence software; decision support software; emergency locator transmitter; image processing system
	Technimeter	185	77.08 %	3d printing; application programming interface; autonomous robot; digital camera; machine learning
	Techopedia	143	39.07 %	actuator; android device; biotechnology; internet of thing; java
	Wikipedia	44	78.57 %	augmented reality; google glass; internet of things; quantum computers; robot
Total Gazetteer-based		449	57.56 %	
Rule-based		883	26.68 %	4g mobile technologies; 5th generation systems; air traffic management systems; cognitive radio network; critical information system

The first column shows the approach used for extracting technologies. The second column shows the number of different collected technologies. The third column shows the precision of the method and the last column contains a random sample of five technologies identified with the respective method.

We used the collected list of entities, aforementioned in Section IV-B, to retrieve the technologies in the patents. For each technology, we create a regular expression (RegEx), namely a sequence of characters that specifies a search pattern [70]. Each RegEx aims at searching technology in the title, abstract, or claim of patents over the 50-year period spanning from 1968 to 2017. The technologies that are classified as similar are used for creating the same regular expression. For example, the RegEx referred to the three-dimensional (*3D*) *printing* technologies is “(\\b3[-]d print*\\b)” in order to include also the different spelling of the *3D printing* as *3D print*, *3D printings*, or *3-D printing*. The similarity and retrieval process was implemented using the R package “stringr” [71].

The ending year of the period is 2017 in order to mitigate the patent filing delay. In fact, when a patent is filed in a patent office, it requires an examination to be granted before its publishing and this period usually lasts 24 months [72].

The technologies are then divided per year based on the filling date of the patent in which the technologies appear. In this way, we built a technological time series of technologies. The output of this phase is a set of lists of technologies per year.

2) *Co-Occurrence Network Building*: For each group of technologies per year, we built a corresponding co-occurrence network calculating the co-occurrence degree of technologies for each year t (i.e., how many times two elements appear together in the same patent). The results are then represented by an (N_b, N_t) adjacency matrix, where N_t is the number of unique technologies. Each matrix contains the number of co-occurrences for each year. We then used the adjacency matrices to generate an undirected network graph $Gst = (Vst, Est)$, where the vertices, Vst , are the technologies and an edge Est exists if two nodes co-occur at least once in the same patent. The weight

¹⁰[Online]. Available: <https://pypi.org/project/spacy/>

of each edge is the respective co-occurrence value between the two vertices.

3) *Community Detection*: After creating a network graph Gst for each year, we used a weighted modularity approach [73] to detect communities. Modularity measures how well each cluster represents the natural subdivision of the nodes based on how strongly connected are the nodes within each group [74]. We choose the subdivision of the nodes that provides the largest modularity. We used the resolution limit of modularity [75] as a parameter-dependent approach to investigate how we could best partition the technologies. Moreover, in Appendix C, we describe the sensitivity analysis we performed to test the robustness of the community detection.

Technology clusters, also called technology communities, are groups of vertices that share common properties and play similar roles within the network. A typical problem of network analysis is picking descriptive human-readable labels for the clusters produced by a clustering algorithm; standard clustering algorithms do not typically produce such labels. A simple cost-effective way of overcoming this limitation is to find the centroid, namely the most important (and representative) node inside a cluster. In the present work, the identification of centroids for each cluster is computed using the *networkx* package and calculating the eigenvalue centrality of each node of the cluster. The eigenvector centrality is a measure of the influence of a node in a network [76]. In this way, we have a label for each cluster in order to give a descriptive fashion to the groups of technologies.

4) *Convergence Patterns Identification*: To identify convergence patterns, we developed a novel approach that detects the dynamic evolution of knowledge. This approach bridges together the assumption that technology convergence is a phenomenon of knowledge evolution [77] with dynamic network analysis [78].

As a starting point, let us note that technologies appear for the first time (birth) and disappear (death). In both cases, a crucial methodological requirement is the ability to distinguish these initial or terminal events from other dynamic patterns in which technologies seem to appear for the first time but originate from a previous technology or seem to disappear but are included in a new one. In other words, in order to detect technological convergence, we need to refer to a knowledge evolution framework in which all dynamic movements of technologies are traced over time and are classified in a consistent way.

Such a framework is offered by Di Caro *et al.* [78], who suggest that the technological evolution may be identified by following the transformation of the vocabulary used in an appropriate corpus of documents related to a domain (see also [48] for an application). This framework conceptualizes five patterns of the dynamic evolution of knowledge: stability, birth, death, merging, and splitting. Previous frameworks proposed in the convergence literature used several patterns (such as knowledge fusion and knowledge combination, see [77]). However, concepts such as stability, birth, death, merging, and splitting are widely used in research works that leverage network analysis to detect the dynamic evolution of communities [79]. These concepts best represent the findings provided by previous approaches that used

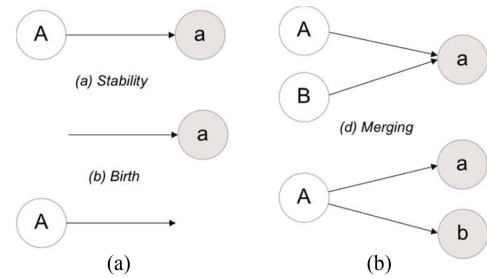


Fig. 1. Patterns of knowledge evolution. (a) *Stability*: a cluster “A” exists at time t and $t+1$. (b) *Birth*: the cluster “a” at time $t+1$ has no antecedent at time t . (c) *Death*: the cluster “A” at time t disappears at time $t+1$. (d) *Merging*: multiple clusters at time t combine in a new topic “a” at time $t+1$. (e) *Splitting*: multiple clusters at time $t+1$ share an antecedent at time t (source: Di Caro *et al.* [78]).

network analysis in patents to detect technological convergence (see Section III-C for references).

The lists of technologies for each year (see Section IV-C1) and their clusters (see Section IV-C3) trace over time the dynamics of technologies. By observing the dynamics of clusters in adjacent time windows, we are able to capture the evolution of the technology and highlight events of birth and death, as well as various types of recombination. At one extreme, we can find a situation in which knowledge does not evolve and, thus, clusters are stable over time. On the other extreme, we figure out a situation of maximum turbulence in which new technologies emerge without any relation with the existing ones. In between these cases, more realistically, we will observe both events of birth and death of technological clusters.

More interestingly, knowledge may evolve by changing its configuration or its clustering structure. Knowledge is reconfigured and recombined, generating new clusters. The new clusters are formed by pre-existing items but with a different configuration. By observing these dynamics, we will be able to identify those events that correspond to the notion of technological convergence, i.e., technologies that become permanently associated and combined together, while in previous periods they were distant.

Fig. 1 summarizes the five ideal patterns of knowledge evolution and their interpretation.

This evolution can be mathematically formalized as follows. Let us consider M clusters emerged as the result of the community detection at time t and N clusters at time $t+1$. We address the critical problem of tracking the transformation of the set of clusters $M = (1, \dots, i, \dots, M)$ at t into the set of clusters $N = (1, \dots, j, \dots, N)$ at $t+1$. Specifically, we are interested in measuring the magnitude of the various phenomena such as stability, birth, death, merging, and splitting. In order to do that, we consider the similarity index $\text{simil}(i, j)$ based on co-occurrence between each couple of clusters (i, j) with $i \in M$ and $j \in N$. We use the Jaccard¹¹ index to calculate the similarity $\text{simil}(i, j)$. The Jaccard index is a similarity metrics that measures the overlap between two sets [80]. In this case, the similarity could be expressed as

¹¹The Jaccard as well as its variation the Szymkiewicz–Simpson Index are the most widely used indexes for similarity calculations between networks.

follows:

$$\text{simil}_{i,j} = js(i, j) = \frac{|i \cap j|}{|i \cup j|} = i \in M, j \in N. \quad (1)$$

We consider the similarity matrix S as the incidence matrix of M over N . We now show how this representation can help measure the magnitude of births, deaths, merging, and splitting.

$$S = \begin{bmatrix} \text{simil}_{11} & \cdots & \text{simil}_{1,N} \\ \vdots & \ddots & \vdots \\ \text{simil}_{M,1} & \cdots & \text{simil}_{M,N} \end{bmatrix}. \quad (2)$$

Births and deaths can be easily calculated from the matrix S . A row sum equals zero highlights a death, while a column sum equals zero indicates a birth. It is important to notice that birth and death are extreme scenarios, while in the reality, we observe a continuum between these two cases. We might, thus, consider the calculation of a *Novelty Index* (NI) following the approaches by previous researchers [81], [82]. We can assume that the sum of the elements of the S matrix quantifies the overall technological similarity. We define an NI_i for each cluster i at time t and vice versa, an NI_j for each cluster j at time $t+I$. For $NI_i = 1$, we have a death, while for $NI_j = 1$ we have a birth. We can also measure an average change in NI by looking at distributions of these indexes over the clusters

$$\begin{cases} NI_i = 1 - \frac{\sum_j^N S_{i,j}}{\sum_i^N S_{i,j}} \\ NI_j = 1 - \frac{\sum_i^M S_{i,j}}{\sum_i^M \sum_j^N S_{i,j}} \\ NI = 1 - \frac{\sum_i^M \sum_j^N S_{i,j}}{M \cdot N} \end{cases}. \quad (3)$$

The interpretation of NI is different according to these three declinations.

First, higher values of NI_i could suggest a counterintuitive interpretation. A cluster i that has low similarity with clusters in the period $t+I$ has high NI_i and, consequently, is more likely to represent a death. In other words, technologies of cluster i with high novelty are more likely to disappear in the period $t+I$. Low values of NI_i , on the contrary, suggest that in the subsequent period, it is possible to find almost all its technology elements.

Second, higher values of NI_j suggest that the cluster j has a small intersection with clusters at the previous period, and thus, it is composed of new emergent technologies. On the opposite, small values of NI_j suggest that the technology components cluster j are more likely to be found also in the previous period.

Finally, the average change in NI could be interpreted as follows: large values of novelty mean there is only a small intersection between the clusters and, consequently, that new technologies are emerging. On the opposite, small values indicate that between two-time windows (t and $t+I$), a small number of technologies emerged. However, these values are not informative enough about the technological turbulence. Novelty can only quantify technological emergence but not convergence. For example, small values of novelty do not exclude the fact that clusters are changing their compositions, and thus, they are repurposing by merging or splitting their technological composition. Due to these limitations of the NI , we need to further explore the similarity matrix.

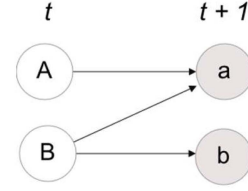


Fig. 2. Bipartite network between clusters at time t {A; B} and clusters at time $t+I$ {a; b}.

Let us consider the following example of simple knowledge evolution between the time window of t and $t+I$: two clusters (A and B) merge in a and one cluster (B) splits into a and b .

Fig. 2 represents the bipartite network of this explanatory case. Consequently, we have the following Similarity matrix S :

$$S = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \quad (4)$$

where α , γ , β , and δ represent the similarity values. In this specific case, α , γ , $\delta \neq 0$, while $\beta = 0$. As mentioned, the transformation of clusters can take the form of merging and splitting. To analyze the intensity of the merging and splitting, we can perform the following matrix multiplications: $S \times S^T$ resulting in the matrix P^{merging} and $S^T \times S$ resulting in the matrix $P^{\text{splitting}}$.

$$\begin{aligned} p_{\text{merging}} &= S \times S^T = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \times \begin{bmatrix} \alpha & \gamma \\ \beta & \delta \end{bmatrix} \\ &= \begin{bmatrix} \alpha \cdot \alpha + \beta \cdot \beta & \alpha \cdot \gamma + \beta \cdot \delta \\ \alpha \cdot \gamma + \beta \cdot \delta & \gamma \cdot \gamma + \delta \cdot \delta \end{bmatrix} \end{aligned} \quad (5)$$

$$\begin{aligned} p_{\text{splitting}} &= S^T \times S = \begin{bmatrix} \alpha & \gamma \\ \beta & \delta \end{bmatrix} \times \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \\ &= \begin{bmatrix} \alpha \cdot \alpha + \gamma \cdot \gamma & \alpha \cdot \beta + \gamma \cdot \delta \\ \alpha \cdot \beta + \gamma \cdot \delta & \beta \cdot \beta + \delta \cdot \delta \end{bmatrix}. \end{aligned} \quad (6)$$

These matrix formulations are useful for computing the intensity of merging and splitting. Given that both matrices P^{merging} and $P^{\text{splitting}}$ are symmetric, we can focus alternatively on the lower or upper triangle. The merging is captured by the value outside the diagonal of the P^{merging} ($\alpha \cdot \gamma + \beta \cdot \delta$). In particular, ($\alpha \cdot \gamma$) is the intensity of the merging of A and B in a , while ($\beta \cdot \delta$) is the intensity of the merging of A and B in b . The splitting is captured by the value outside the diagonal of the $P^{\text{splitting}}$ ($\alpha \cdot \beta + \gamma \cdot \delta$). In particular, ($\alpha \cdot \beta$) is the intensity of the splitting of A in a and b , while ($\gamma \cdot \delta$) is the intensity of the splitting of B in a and b .

As we can see, the elements above (or under) the diagonal of these matrices represent respectively the intensity of merging and splitting. For this reason, we can formulate a *merging index* (MI) and a *splitting index* (SI) as follows:

$$MI = \sum_{i>j} p_{\text{merging}}(i, j) \quad (7)$$

$$SI = \sum_{i>j} p_{\text{splitting}}(i, j). \quad (8)$$

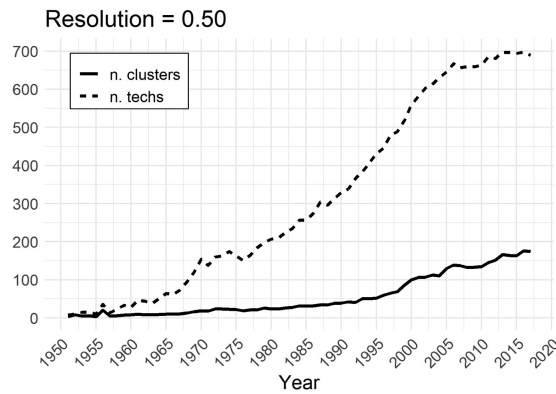


Fig. 3. Trends of the number of technologies in patents and their respective clusters in the C4ISTAR field from 1950 to 2017. The clusters of technologies have been identified choosing the largest modularity that resulted in a resolution value of 0.5 (see Appendix C).

Since the similarity index (Jaccard coefficient) is bounded between 0 and 1, NI, MI, and SI range from 0 to 1.

V. RESULTS

In this section, we show the results obtained using the methodology described in Section IV with a focus on the outcome of the collection of C4ISTAR technologies and the patterns of convergence in the technological landscape.

A. Results of Technologies Collection

The text mining techniques applied in this article allow us to recognize 1090 technologies from the C4ISTAR documents. Table III provides the number of technologies identified and the precision for each NER approach. The last column of Table III shows a sample of technologies collected using the various methods. As we can see, in some cases, the different approaches identify the same technology (242 technologies were detected by both gazetteer and rule-based approaches).

Table III tabulates that the rule-based approach identifies more technologies but is less precise than the gazetteer-based approach (26.68% versus 57.56%, respectively). As discussed in Section III-B, few studies exist on technology Named Entity Recognition. This makes it hard to compare the results to other works. It is important to note that the precision does not impact the quality of the output (effectiveness of the whole process) but only the time needed to revise the final list of technologies (efficiency, which decreases due to manual review). For this reason, the fact that more than 2/3 of the technologies (883 on 1090 technologies) come from the rule-based approach with very low precision has no impact or limitation on the process.

B. Results of Technological Convergence Analysis

In this section, we show the results of the technological convergence analysis.

Fig. 3 shows the distribution of the technologies and their clusters over the years. The fact that technologies are growing over the years is consistent with previous research on technological

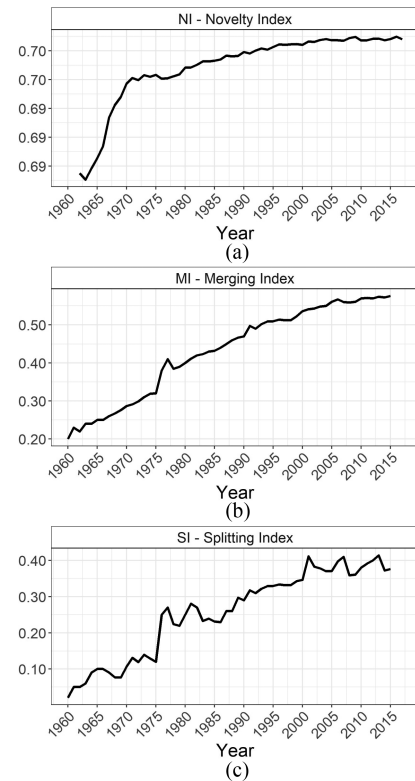


Fig. 4. Trends of (a) NI, (b) merging index, and (c) splitting index calculated from the technology clusters extracted from 1960 to 2017.

evolution [59] and in particular on the C4ISTAR field. In fact, as the convergence of scientific knowledge coming from different disciplines becomes more and more pervasive, the emergence of new technologies becomes more frequent [57].

Fig. 4 shows the measures of knowledge evolution, such as birth, death, stability, (a) merging, (b) splitting, and (c) using our combined text-dynamic network methodology.

The time series illustrated in Figs. 3 and 4 lead us to several considerations. First, we find an impressive growth in the number of distinct technologies employed in the defense sector. We also find remarkable growth in the number of clusters of technologies. One may wonder whether this is a nominal effect, driven by the labeling of technologies over time. Given the precision of the profiling methodology, we tend to exclude a purely linguistic effect. What we see is a powerful process of attraction of many nonmilitary technologies in the defense domain of application.

Second, the NI shows a jump in the 1960s and increases almost monotonically afterward. It would be interesting to investigate in detail the reasons underlying this pattern. We put forward the conjecture that the C4ISTAR field started to incorporate electronics in the 1960s and other computer-intensive technologies afterward.

Third, we see a parallel growth of the Merging and Splitting indexes. This suggests that these technological trends are complementary in generating the dynamics of recombination.

Fourth, while the trend is similar, the Merging index is systematically much larger than the Splitting index. We see clear confirmation of the process of technological convergence. The

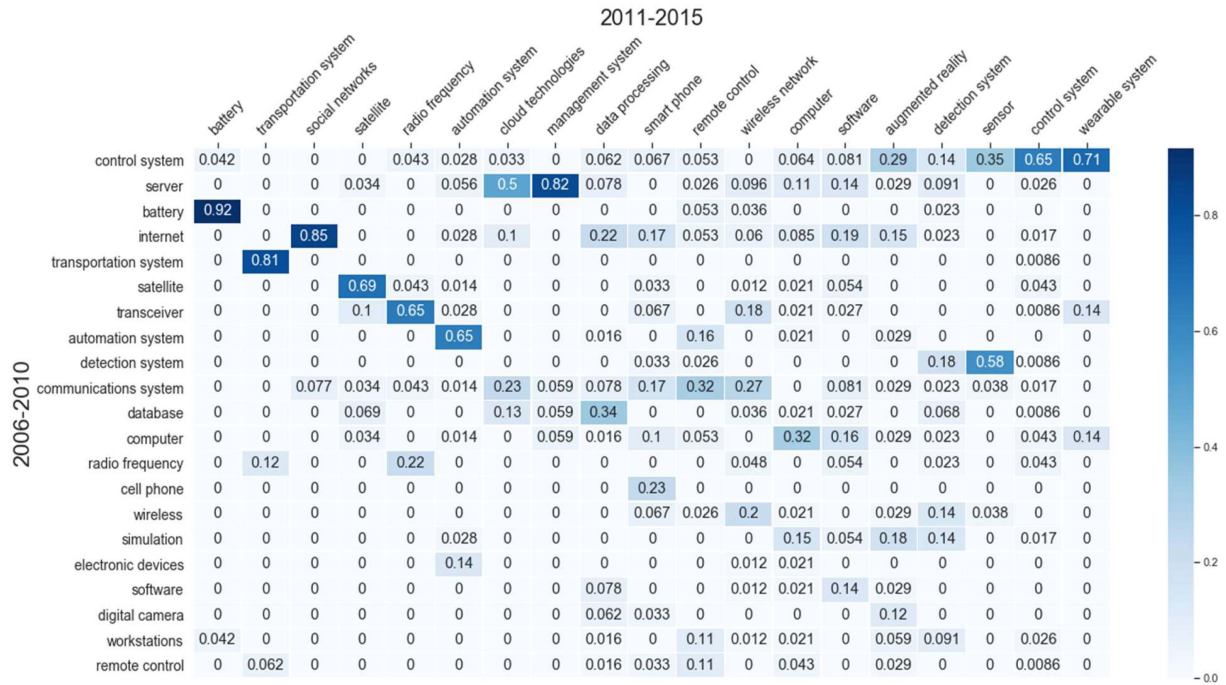


Fig. 5. Heatmap of the similarity matrix between clusters identified in the 2006–2010 period (21 clusters on the rows) and the 2011–2015 period (19 clusters on the columns). Each value of the matrix is calculated by computing the Jaccard index. The higher is the similarity between the clusters, the darker (blue) is the cell. Value cells with 0 represent no similarity between the clusters, namely they do not share technologies. Each cluster is labeled with its centroid.

growth in the number of distinct technologies and clusters, illustrated in Fig. 3, is accompanied by a turbulent dynamic, made visible by splitting and merging in Fig. 4. This turbulent dynamic is remarkable since the field of C4ISTAR technologies has extremely long development project timelines and relatively slow adoption of radically new technologies.

Merging is, however, prevalent, suggesting a dynamic in which independent technologies are used together for the resolution of complex challenges and then create new clusters, co-occurring on a regular basis on documents. This is consistent with the notion of systemic innovation suggested by some authors.

As already discussed, several authors have suggested that technology convergence is rising [57]. In most cases, however, the empirical basis for this statement has been based on classifications of patents or publications or on keywords. It is important to underline that our results, based on the theory of Di Caro *et al.* [78], give text-based evidence of this phenomenon.

On the basis of this novel evidence, we suggest that the technological convergence in the defense sector has some resemblance with the convergence observed in the so-called Fourth Industrial Revolution, or Industry 4.0. It has been argued [22] that this transformation is not a radical innovation per se, but a recombination of existing introduced technologies based on cyber-physical integration. Therefore, we are interested in analyzing if such recombination of technologies affects also the C4ISTAR domain.

An emergent technological paradigm could cover a relatively long-time span, such as a decade and beyond [83]. For this reason, we analyze the evolution of technological patterns for the most recent periods: 2006–2010 and 2011–2015. Fig. 5 shows

the similarity matrix between the clusters of 2006–2010 and those of 2011–2015.

A closer discussion of Fig. 5 helps us to understand how the spread of turbulence introduced by the Fourth Industrial revolution affected the C4ISTAR domain.

This similarity matrix shows a sharp difference in the dynamic patterns. In some cases, the similarity between clusters at different dates is very high (dark blue in the figure). In these cases, we expect stability of the technology over time. In other cases, on the contrary, clusters identified in 2006–2010 period are weakly similar to clusters with the same centroid in the subsequent period. They appear more similar to several other clusters. A more detailed view of the underlying dynamics is offered by Table D of Appendix D. This table provides samples of the technologies shared by clusters whose similarity values are greater than or equal to 0.2.

Moreover, it is possible to observe cases that confirm the recombination scenarios we conceptualize. On the one side, clusters such as “control system” or “server” (from the initial period) show an intense splitting. On the other side, clusters such as “sensors” or “remote control” (from the final period) remark a consistent merging.

For what concerns the novelty of the first period [see Fig. 6(a)], we can see that values are on average high (mean of *NI* in the first period equals 0.67). A high *NI* in the first period could be caused by the fact that technologies are disappearing. This plot, in fact, needs to be interpreted with the help of Fig. 5. For example, the cluster “workstation” has the highest *NI* and, intuitively, has low similarity values with other clusters in the latter period. This means that technologies belonging to this cluster disappeared. On the opposite, the cluster “battery” has the lowest novelty due

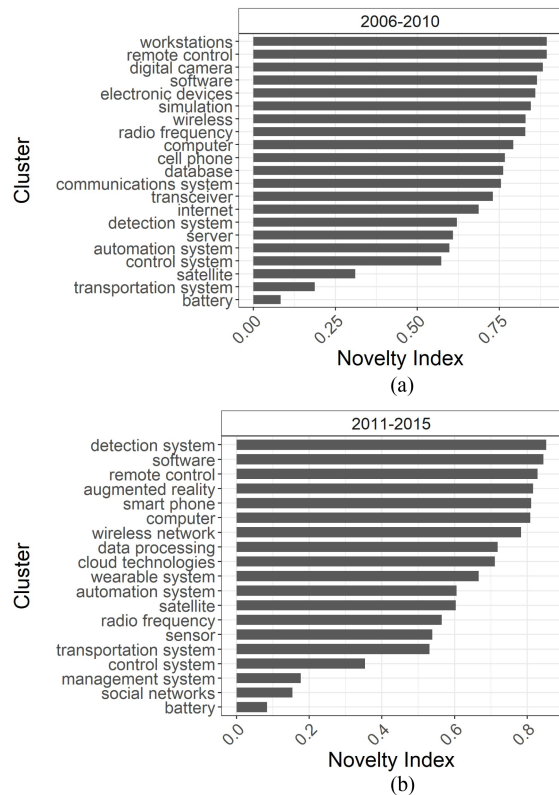


Fig. 6. Barplots of NI for the technology clusters of the two periods: (a) 2006–2010 and (b) 2011–2015.

to the fact that shares almost all of its technologies with the latter period.

However, the high disappearance of technologies is accompanied by an even more growing appearance of technologies in the subsequent plot in Fig. 6(b). The average *NI* is higher in the second period (it equals 0.71). This means that the number of technologies has grown.

Let us consider two extreme cases in order to further understand the interpretation of these results. On one extreme, clusters such as “remote control” or “software” have high novelty in both periods. This means that these two clusters, even though they did not change their centroids, showed high turbulence, not in terms of merging and splitting, but in terms of births and deaths: a high number of technologies disappeared and have been replaced by another higher number of new ones. On the other side, it is clear the behavior of clusters such as “battery” or “transportation systems” whose low values of novelty indicate the stability of technologies within it.

Another important consideration regards those clusters that have high similarity but different centroids. This is the case, for example, of “server” that splits into “cloud technology” and “management systems.” This example shows that a part of the “server” cluster emerged in the combination with the elements of other clusters. However, its high similarity with “management systems” suggests that its remaining technologies that did not converge in other clusters found a repurposing in terms of centroids. This particular dynamic, as well as other ones, such as “social network” that shares its technology mostly with “Internet,” or the “transceiver” that converged into “radio

frequency” shows that technological convergence that we are attempting to capture does not occur only by means of birth, death, merging and splitting: the change of centroids could also suggest a repurposing.

C. Discussion of the Results

The results show a remarkably high level of technological convergence. This may be due to various factors. On the one hand, the defense sector has been subjected, with a delay with respect to other industries, to an intense acquisition of IT solutions that have revolutionized the architecture of traditional mission systems. Data acquisition, data processing, strategic and operational intelligence, remote control, landscape acquisition, and reconnaissance are all examples of the integration of IT into conventional warfare systems. Our data give large evidence of the massive impact of this type of convergence at the macrolevel. Based on our analysis, it would be possible to examine at a more granular level individual technologies and/or solutions and applications.

On the other hand, it is possible that the convergence is affected by the effort to use civilian technologies to cut development costs [9], as well as the effort to generate civilian applications for defense technologies. In order to give evidence of the magnitude of this effect, it would be possible to examine the corporate origins of patents (i.e., military versus nonmilitary assignees) as well as to classify patents in terms of functions, or user groups, or advantages and disadvantages [52].

The data also show that the main drivers of technological evolution lie in the two kinds of dynamics: large turbulence of entry of new technologies and progressive acquisition and integration of new technologies into existing competencies.

VI. CONCLUSION AND FUTURE RESEARCH

This article has some limitations. The first and more obvious is that it is based on patent analysis. The limitations of patents in the analysis of innovation are well known and largely discussed in the literature [84]. In the military sector, it is possible that some important innovations are not patented at all but are kept secret. At the same time, given the increasing knowledge flows between the military and the civil sectors [85], it is reasonable to assume that a large share of technologies of interest to the C4ISTAR field is appropriately represented in patents.

Another limitation is that the data do not allow to observe changes in extra-technological considerations, such as the geopolitical landscape or the military doctrines. During the half-century covered by our data, there have been major political and military events (from the fall of the Berlin Wall to September 11, to name a few), which are imperfectly recorded in our data.

Third, it would be useful to investigate the technological convergence using a qualitative case study approach, going in depth into the understanding of the role of technologies in specific applications.

Finally, there is room for further methodological improvements. Our methodology still makes use of expert advice (although for a very limited portion of the workflow) and is based on several choices regarding the queries and the technology collection. In further research, it would be important to examine

the robustness of the results by experimenting with alternative structures. Also, we documented that the utilization of advanced NER and NLP techniques, such as BERT-based languages, is inhibited by the lack of BERT languages dedicated to technologies. It is likely that further developments will be soon available in this direction.

At the same time, the methodology developed in this article opens the way to several promising research directions. The first one is the detection of dual-use technologies and the analysis of the rate of growth of technologies in the military and civil areas. Dual-use technologies are defined as those developed and used by the military or space sector and by the civilian sector [86]. The second area of analysis deals with the foresight of emerging technologies in the C4ISTAR domain, based on an understanding of their growth pattern [87]. In this direction, another promising investigation is the relationship between technological applications and scientific research. Many authors have observed that the interactions and interdependencies between science and technology have been increasing over time. At the same time, large gaps still exist between them [88]. The combined analysis of scientific papers and patents in the C4ISTAR domain can be used to determine the gaps and identify the future trends of emerging technologies.

Summing up, this article offers both methodological and substantive contributions to the issues of technology convergence and long-term technological and strategic evolution of the complex and systemic domain of defense. Our novel methodologies have a broad range of future applications. In turn, empirical analyses of the defense sector might start from the rich repository of data produced by the application of the methodology in order to qualify, with quantitative or qualitative approaches, the interpretation of the patterns clearly identified in this article.

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APPENDIX

Supplementary material associated with this article can be found, in the online version of the Appendix document, at <https://data.mendeley.com/datasets/8wj4kdmhc4/1>.

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